Harsh Gharewal

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Education

Shri Govindram Seksaria Institute of Technology and Science — Indore, Madhya Pradesh
Bachelor of Technology in Mechanical Engineering — *Sept. 2023 – May 2027*

Experience

Clustal — June 2026 – July 2026
SDE Intern — *Indore, Madhya Pradesh*

- Built and optimized responsive web applications using React.js and Next.js, delivering fast, scalable, and user-friendly frontend experiences across desktop, tablet, and mobile devices.
- Converted Figma designs into pixel-perfect interfaces while enhancing website visuals through AI-assisted image generation, image quality optimization, engaging scroll animations, and custom loading screens.
- Utilized Git for collaborative version control and Vercel for deployment, following modern frontend development practices to ensure maintainable, high-performance applications.

Indian Institute of Technology Jammu — April 2026 – Present
Research Intern — *Remote*

- Engineered REACH, an RL-based microservice scheduler using PPO achieving 7.9-10x latency reduction through adaptive pod rescheduling on Kubernetes
- Built simulation-to-real cloud-edge optimization framework, achieving 10x latency reduction and 86.9 request stability under dynamic conditions.

Engineer Sahab — June 2025 – July 2025
Web Development Intern — *Indore, Madhya Pradesh*

- Developed clean, responsive web interfaces using HTML, CSS, JavaScript/jQuery, and Bootstrap
- Implemented interactive UI components like carousels, animations, and dynamic menus.
- Contributed 50K+ lines of code to an established codebase via Git.
- Enhanced website responsiveness and usability, improving performance by up to 70x.
- Collaborated with designers/back-end developers to translate UI/UX wireframes into functional web pages.
- Debugged and fixed UI issues, improved layout consistency, and refactored front-end code for better maintainability.

Projects

JEE RAG | *Python, Flask, React, PostgreSQL, Docker* — July 2026 – Present

- Built a RAG API with FastAPI, Groq (LLaMA 3.3 70B), Qdrant vector DB, and Tavily web search for solving JEE Advanced problems in Physics, Chemistry, and Math.
- Implemented hybrid retrieval combining dense cosine similarity with BM25 scoring (0.6/0.4 weighted fusion) for improved context relevance.
- Designed an async pipeline with concurrent Ollama embeddings, overlapping text chunking, and structured prompt engineering with subject-specific reasoning rules.

Technical Skills

Languages: C, C++, JavaScript, Python
Frontend: ReactJs, Tailwind, HTML, CSS, UI Component Libraries
Backend: NodeJs, Express.js, API Integration
GenAI Development: LangChain, LangGraph, RAG Pipelines, Vector Databases,Agentic AI, MCP Servers
Databases and Tools: MongoBD, Git/GitHub, GitHub Actions

SCHOLASTIC ACHIEVEMENTS

- Participated in various hackathon like Hackwithinfy, Google Big Code 2026, GS India Hackathon etc
- Graffathon — Golden Book of Records 2025 Led "Graffathon, " a 33-hour wall painting competition with 3500+ national participants; secured sponsorships and set a world record in the Golden Book of Records 2025.
- Streetscape — World Book of Records 2024 Organized "Streetscape, " a 30-hour road painting competition with 3000+ participants; secured sponsorships and set a world record in the World Book of Records 2024.